

A Generative AI and RAG Based Intelligent System for Campus Problem Detection and Resolution

*A Project Report Submitted in the
Partial Fulfillment of the Requirements
for the Award of the Degree of*

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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Acknowledgements

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **Mr. Ganesh Deshmukh**, Assistant Professor and Project Supervisor, Department of Computer Science and Engineering, Vardhaman College of Engineering, for his able guidance and useful suggestions, which helped us in completing the project in time.

We are particularly thankful to **Dr. Gagandeep Arora**, the Head of the Department, Department of Computer Science and Engineering, his guidance, intense support and encouragement, which helped us to mould our project into a successful one.

We show gratitude to our honorable Principal **Dr. J.V.R. Ravindra**, for providing all facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartfelt thanks to **Dr. Teegala Vijender Reddy**, Chairman and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing a congenial atmosphere to complete this project successfully.

We also thank all the staff members of Computer Science and Engineering department for their valuable support and generous advice. Finally thanks to all our friends and family members for their continuous support and enthusiastic help.

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Abstract

Infrastructure and maintenance issues such as broken laboratory equipment, electrical faults, water leakages, damaged furniture, and sanitation problems are common in educational institutions and negatively impact the learning environment, safety, and overall campus experience. Existing systems for reporting and managing such issues are often manual, slow, and unorganized, leading to delays, lack of transparency, and inefficient resource utilization. To address these challenges, this project proposes an intelligent system for automatic problem detection and resolution in a smart campus environment using Generative Artificial Intelligence (AI) and Retrieval-Augmented Generation (RAG). Users such as students, faculty, and staff can report issues through a structured interface, where the system analyzes the input, classifies the type of problem (e.g., electrical, plumbing, infrastructure), and assigns priority levels (High, Medium, Low) based on severity. The RAG approach enhances system reliability by retrieving relevant information from a structured knowledge base, including maintenance records, past complaints, policies, and standard procedures, and combining it with AI-generated responses. The system also incorporates a learning mechanism that improves performance over time using historical data and user feedback. Additionally, it ensures transparency by allowing users to track complaint status, while administrators can monitor issues and optimize resource management. Overall, the proposed system provides a smart, scalable, and efficient solution for campus problem management, reducing manual effort, improving response time, and enhancing the overall campus experience.

Keywords: Smart Campus; Generative Artificial Intelligence(AI); Retrieval-Augmented Generation (RAG); Automated Problem Detection

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Abbreviations

Abbreviation	Description
VCE	Vardhaman College of Engineering
RAG	Retrieval-Augmented Generation
SOP	Standard Operating Procedures

CHAPTER 1

Introduction

1.1 Introduction

Infrastructure and maintenance-related issues such as broken laboratory equipment, electrical faults, water leakages, damaged furniture, and sanitation problems are commonly observed in educational institutions. These issues disrupt academic activities and affect the safety, hygiene, and overall campus experience of students and staff. Existing systems for reporting and resolving such problems are often manual, time-consuming, and unstructured, relying on emails, phone calls, or paper-based forms. This leads to delays, lack of proper tracking, and inefficient resource management.

To address these challenges, this project proposes an intelligent smart campus system for automated problem detection and resolution using Generative Artificial Intelligence (AI) and Retrieval-Augmented Generation (RAG). The system enables users to report issues through a structured interface, where the AI analyzes the input, classifies problems into categories such as electrical, plumbing, or infrastructure, and assigns priority levels based on severity. The integration of RAG enhances system reliability by combining AI-generated responses with relevant institutional knowledge, including maintenance records, policies, and past issue data, resulting in accurate and context-aware outputs.

The project aims to develop a scalable and efficient platform that improves transparency and optimizes resource utilization in campus maintenance systems. It covers key functionalities such as problem reporting, classification, prioritization, response generation, and continuous learning from past data. The system is adaptable to different institutions and supports administrators in monitoring issues and planning maintenance activities effectively.

1.2 Background and Motivation

The project is situated in the field of smart campus systems, which focuses on using modern technologies such as Artificial Intelligence (AI) and data-driven solutions to improve campus operations and management. A key area within this field is the efficient handling of infrastructure and maintenance issues, including electrical faults, water leakages, damaged equipment, and sanitation problems. These issues require timely resolution to ensure a safe, hygienic, and productive environment in educational institutions.

Traditionally, such problems are managed through manual or semi-digital methods like complaint registers, emails, or basic web forms. Although these methods allow issue reporting, they lack automation, proper tracking, and intelligent decision-making, leading to delays, poor prioritization, and inefficient resource utilization. Recent approaches have introduced machine learning for classification, IoT-based monitoring, and generative AI for understanding user input. However, standalone generative models may produce inaccurate outputs due to limited domain knowledge, which is addressed by Retrieval-Augmented Generation (RAG) that combines AI with structured data retrieval for improved accuracy.

Despite these advancements, there is still a lack of unified systems that integrate reporting, classification, prioritization, and resolution specifically for smart campuses while effectively utilizing institutional knowledge such as past records and policies. This project aims to bridge this gap by developing an intelligent system using Generative AI and RAG to automate problem detection and resolution. The proposed system enhances efficiency, reduces manual effort, ensures faster responses, and improves transparency, contributing to a smarter and more efficient campus environment.

1.3 Objectives of the Project Work

The main aim of this project is to develop an intelligent and automated system for detecting, classifying, and resolving infrastructure-related issues in a smart campus environment using Generative Artificial Intelligence (AI) and

Retrieval-Augmented Generation (RAG). The objectives of this project are as follows:

1. To develop a user-friendly system that allows students, faculty, and staff to report campus-related problems in a structured format.
2. To design and implement an AI-based model that can analyze user inputs (text and images) to understand and interpret the reported issues.
3. To classify the reported problems into appropriate categories such as electrical, plumbing, infrastructure, and equipment-related issues.
4. To assign priority levels to reported issues based on their severity and impact to ensure timely resolution.
5. To integrate a Retrieval-Augmented Generation (RAG) mechanism that retrieves relevant institutional data such as maintenance records, policies, and past complaints.
6. To generate accurate and context-aware responses by combining generative AI capabilities with retrieved knowledge.
7. To develop a system that can learn from previously resolved issues and improve its performance over time.
8. To provide transparency by enabling users to track the status of their complaints and receive updates.
9. To assist administrators in efficient resource allocation and decision-making through data-driven insights.
10. To build a scalable and efficient smart campus solution that improves response time, reduces manual effort, and enhances overall campus management.

1.4 Organization of the Report

This report is organized into several chapters to provide a clear and structured understanding of the proposed smart campus problem detection and resolution system.

Chapter 2: Literature Survey presents an overview of existing research and related work in the areas of smart campus systems, artificial intelligence, and automated problem management. It highlights the strengths and limitations of current approaches.

Chapter 3: System Analysis focuses on understanding the existing system and its drawbacks. It also explains the requirements of the proposed system, including functional and non-functional aspects.

Chapter 4: System Design describes the overall architecture of the proposed system. It includes details about system components, data flow, and design models used for implementation.

Chapter 5: System Implementation explains the technologies, tools, and methods used to develop the system. It provides details about how different modules are implemented.

Chapter 6: System Testing discusses the testing strategies used to evaluate the system. It includes different types of testing performed to ensure accuracy, reliability, and performance.

Chapter 7: Results and Discussion presents the outcomes of the implemented system. It analyzes the performance, effectiveness, and improvements achieved through the proposed solution.

Chapter 8: Conclusions and Future Scope summarizes the overall work, key findings, and contributions of the project. It also suggests possible enhancements and future developments for improving the system further.

CHAPTER 2

Literature Survey

2.1 Introduction

The literature survey plays an important role in this project by providing an understanding of existing research and technologies related to smart campus systems, artificial intelligence, and automated problem management. It focuses on reviewing previous works that address infrastructure maintenance, issue reporting systems, and intelligent automation. This helps in identifying current trends, strengths, and limitations in existing approaches.

The methodology involves analyzing research papers, articles, and existing systems related to AI-based problem detection, Generative AI, and Retrieval-Augmented Generation (RAG). Relevant sources are selected based on their contribution and applicability to the proposed system. The review mainly examines how different technologies are used for problem classification, response generation, and overall system automation.

Studying prior research helps in identifying gaps in existing systems, such as lack of automation, improper prioritization, and limited intelligent decision-making. Although some modern approaches use machine learning and AI, they often fail to provide accurate and context-aware responses due to limited domain knowledge. This highlights the need for advanced techniques like RAG, which combines generative AI with structured data retrieval to improve reliability. Overall, this survey establishes a strong foundation for the proposed system and justifies the need for an efficient and intelligent smart campus solution.

2.2 Review of Prior Research

This section reviews existing research and systems related to smart campus management, automated issue detection, and the use of Artificial Intelligence

(AI) in problem-solving. It covers key areas such as traditional complaint management systems, machine learning-based approaches, IoT-based solutions, and recent advancements in Generative AI and Retrieval-Augmented Generation (RAG). This review helps in understanding current developments and identifying the strengths and limitations of existing solutions.

Traditional complaint management systems in educational institutions rely on manual or semi-digital methods such as paper registers, emails, or basic web portals. While these systems allow users to report issues, they lack automation, real-time tracking, and intelligent decision-making, resulting in delays, poor prioritization, and lack of transparency. Machine learning techniques have been introduced to improve classification of complaints using algorithms like decision trees, support vector machines, and neural networks. Although these methods enhance classification accuracy, they often struggle with contextual understanding, require large labeled datasets, and have limited adaptability.

IoT-based solutions enable real-time detection of issues using sensors for monitoring conditions such as water leakage, temperature, and electrical faults, but they increase cost and are limited to specific problem types. Generative AI has improved the ability to understand user input and generate responses; however, standalone models may produce inaccurate outputs due to lack of domain knowledge. Retrieval-Augmented Generation (RAG) addresses this issue by combining generative AI with structured data retrieval, improving accuracy and relevance. Despite these advancements, there is still a lack of integrated systems that provide a complete end-to-end solution for smart campus problem management.

2.3 Identified Research Gaps

From the review of existing systems, several key gaps have been identified in smart campus problem management solutions. Firstly, most systems lack end-to-end integration, as they focus only on specific tasks like reporting or classification without covering the complete process from detection to resolution. Secondly, there is limited use of intelligent decision-making, as traditional systems do not effectively use AI for understanding and prioritizing issues.

Another major gap is the absence of context-aware responses. Standalone AI models may generate inaccurate outputs due to lack of access to institutional knowledge such as past records and policies. Additionally, many systems do not include learning mechanisms, making them unable to improve over time.

IoT-based solutions are also limited, as they only detect certain physical issues and require additional hardware. Furthermore, there is a lack of transparency, as users often cannot track the status of their complaints.

This project addresses these gaps by providing an integrated system using Generative AI and RAG, enabling intelligent, accurate, and efficient problem detection and resolution in a smart campus environment.

2.4 Summary

This chapter presented a review of existing research related to smart campus systems, automated problem management, and the use of Artificial Intelligence (AI) in issue detection and resolution. It highlighted how traditional systems are mostly manual and lack efficiency, while modern approaches using machine learning and IoT provide partial improvements but still have several limitations.

The review also discussed the role of Generative AI and Retrieval-Augmented Generation (RAG) in improving problem understanding and response generation. While these technologies show significant potential, their implementation in campus management systems is still limited.

Several research gaps were identified, including the lack of end-to-end integrated systems, limited use of intelligent decision-making, absence of context-aware responses, and lack of continuous learning and transparency. These gaps indicate the need for a more advanced and unified solution.

Based on these findings, this project aims to develop an intelligent system that combines Generative AI and RAG to provide accurate, efficient, and scalable problem detection and resolution. This work builds upon existing research and addresses the identified gaps, contributing to the development of a smarter and more effective campus management system.

CHAPTER 3

System Analysis

3.1 Existing System

The existing system for handling infrastructure and maintenance issues in educational institutions is mostly manual or semi-digital. In many colleges and universities, problems are reported through methods such as complaint registers, emails, phone calls, or basic web forms. These systems allow users to report issues, but they lack proper structure, automation, and intelligent processing.

In manual systems, complaints are written in registers or communicated verbally, which makes it difficult to track the status of issues. There is no proper record management, and complaints may be overlooked or delayed. In semi-digital systems, although complaints can be submitted online, they are still processed manually by administrators, which leads to slow response times and inefficient handling.

Another limitation of the existing system is the lack of automatic classification and prioritization. All complaints are treated equally, without considering the severity or urgency of the issue. This may result in critical problems being delayed while less important issues are addressed first.

Additionally, existing systems do not provide real-time updates or transparency. Users are often unaware of the status of their complaints, which leads to dissatisfaction and lack of trust in the system. There is also no proper use of historical data or past records to improve decision-making.

Overall, the existing system is time-consuming, inefficient, and not suitable for large-scale campus environments. These limitations highlight the need for an intelligent, automated, and scalable solution.

3.1.1 Disadvantages of Existing System

The existing systems used for managing campus infrastructure issues have several limitations that affect their efficiency and reliability.

- Manual and time-consuming process, as most systems rely on complaint registers, emails, or phone calls.
- Lack of automation, with no intelligent system to analyze, classify, or prioritize complaints.
- Poor tracking mechanism, making it difficult for users to know the status of their complaints.
- No prioritization of issues, leading to delays in resolving critical problems.
- Inefficient resource utilization due to lack of proper analysis and decision-making.
- Lack of data utilization, as historical records are not used for improving system performance.
- Delayed response time caused by manual processing and lack of coordination.
- Limited scalability, making the system unsuitable for large campus environments.

3.2 Proposed System

The proposed system is an intelligent smart campus problem detection and resolution system that uses Generative Artificial Intelligence (AI) and Retrieval-Augmented Generation (RAG) to automate and improve the complaint management process.

In this system, users such as students, faculty, and staff can report infrastructure-related issues through a structured interface by providing a description and optionally uploading images. The system uses AI to analyze the input, understand the problem, and classify it into categories such as electrical, plumbing, infrastructure, or equipment-related issues.

The system also assigns priority levels to the reported issues based on their severity and impact, ensuring that critical problems are addressed quickly. A key feature of the proposed system is the integration of Retrieval-Augmented Generation (RAG), which retrieves relevant information from institutional data sources such as maintenance records, policies, and past complaints.

By combining retrieved knowledge with generative AI, the system generates accurate and context-aware responses. It also includes a learning mechanism that improves performance over time by analyzing previously resolved issues.

Additionally, the system provides transparency by allowing users to track the status of their complaints and receive updates. Administrators can use the system to manage resources efficiently and make better decisions.

Overall, the proposed system offers a scalable, efficient, and intelligent solution for automated problem detection and resolution in a smart campus environment.

3.2.1 Advantages of Proposed System

The proposed system offers several advantages over the existing system by improving efficiency, accuracy, and overall management of campus issues.

- Automated problem detection and classification using Artificial Intelligence.
- Faster response time due to intelligent prioritization of issues.

- Accurate and context-aware responses using Retrieval-Augmented Generation (RAG).
- Improved transparency with real-time tracking of complaint status.
- Efficient resource utilization through better decision-making and analysis.
- Continuous learning capability that improves system performance over time.
- Reduced manual effort and minimized human errors.
- Scalable system suitable for large campus environments.
- Enhanced user experience with a structured and user-friendly interface.

3.3 System Requirements

The system requirements define the necessary hardware and software components needed to develop and run the proposed smart campus problem detection and resolution system.

3.3.1 Hardware Requirements

1. **Processor:** Intel i5 or above (or equivalent)
2. **RAM:** Minimum 8 GB (16 GB recommended)
3. **Storage:** At least 20 GB free space
4. **System Type:** 64-bit system
5. **Internet Connection:** Required for cloud access (Google Colab, Firebase, GitHub)
6. **GPU (Optional):** For faster model training in Google Colab

3.3.2 Software Requirements

1. **Programming Language:** Python
2. **Development Environment:** Google Colab, VS Code or PyCharm
3. **Backend Technologies:** Python Standard Libraries (`http.server`, `json`, `os`, `re`, `urllib`)
4. **Custom Modules:** `ai_engine`, `gemini_service`, `rag_pipeline`, `chatbot_engine`
5. **Frontend Technologies:** HTML, CSS, JavaScript
6. **Cloud and Storage:** Google Colab, Google Drive, Firebase
7. **Version Control:** Git, GitHub
8. **Operating System:** Windows, Linux, or macOS

CHAPTER 4

System Design

4.1 System Architecture

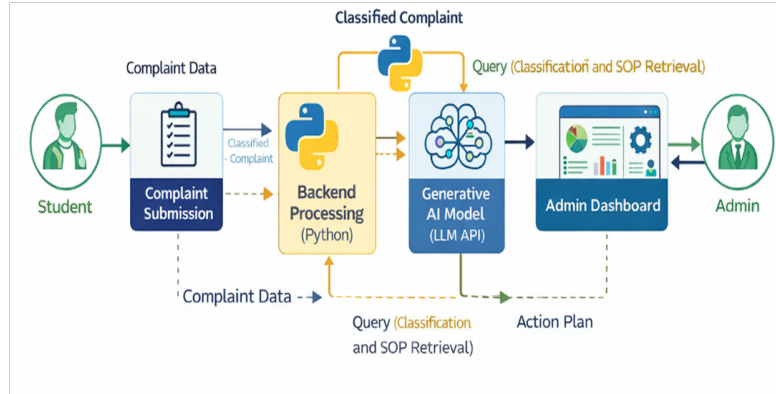


Figure 4.1: System Architecture Diagram.

The system architecture represents the workflow of the smart campus problem detection and resolution system using Generative AI and Retrieval-Augmented Generation (RAG).

- **Complaint Submission:** The process begins with the student or user submitting a complaint through the interface, providing a description of the issue.
- **Backend Processing:** The complaint data is received by the backend (Python), where it is preprocessed and prepared for further analysis.
- **Generative AI Model:** The processed input is sent to the Generative AI model (LLM API), which understands the complaint, classifies the issue, and generates a suitable response.
- **RAG-based Retrieval:** The system retrieves relevant information such as Standard Operating Procedures (SOPs) and past records to enhance the accuracy and context of the response.

- **Complaint Classification:** The system automatically categorizes the complaint into types such as electrical, plumbing, or infrastructure.
- **Admin Dashboard:** The classified complaint along with the generated response is displayed on the admin dashboard for further action.
- **Action Plan:** The system provides a suggested action plan, helping administrators to resolve the issue efficiently.
- **Data Storage and Learning:** The complaint data is stored for future reference, enabling the system to improve performance over time.

4.2 Data Flow Diagram

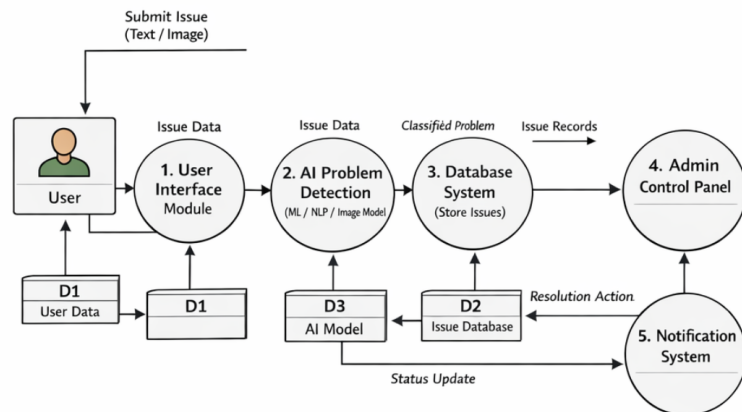


Figure 4.2: Data Flow Diagram

This Data Flow Diagram illustrates how the Smart Campus AI system processes user-reported issues efficiently. Users submit problems through the interface, which are analyzed by the AI module for classification. The system stores and forwards these issues to the admin for resolution. Finally, the notification system updates users with the status or solution, ensuring a complete feedback loop.

4.3 UML Diagrams

4.3.1 Class Diagram

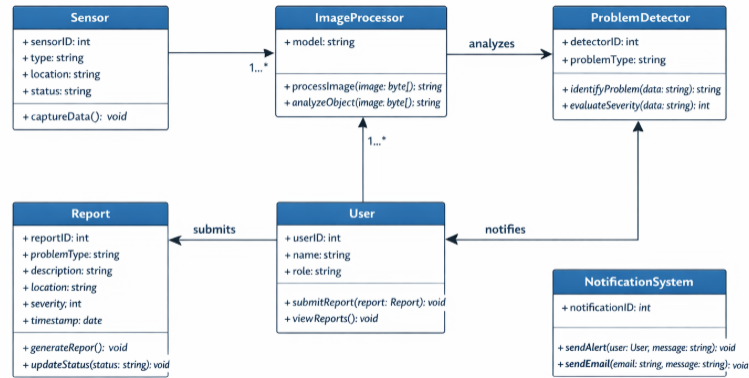


Figure 4.3: Class Diagram

This class diagram represents the structure of the Smart Campus Problem Detection System, showing key components like User, Report, AI Module, and Notification System. It illustrates how users submit issues, which are processed by the AI module and stored in the system. The admin manages and resolves these issues, while the notification system keeps users updated on the status.

4.3.2 Object Diagram

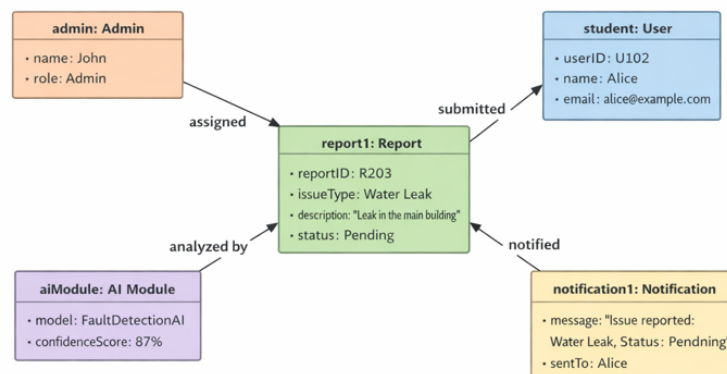


Figure 4.4: Object Diagram

This object diagram represents a real-time snapshot of the Smart Campus Problem Detection System, showing how different objects like user, admin, report, AI module, and notification system interact at a specific moment. It demonstrates how a user submits an issue, which is then analyzed by the AI module and assigned to the admin for resolution. The notification system updates the user about the status, helping to clearly understand the system’s dynamic behavior.

4.3.3 Collaboration Diagram

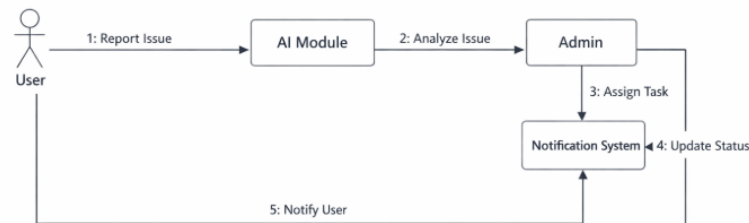


Figure 4.5: Collaboration Diagram

A collaboration diagram shows how different parts of a system communicate with each other to complete a task. It highlights the flow of messages between components like user, AI module, admin, and notification system. In this project, it explains how an issue is reported, processed, resolved, and communicated back to the user.

4.3.4 Use Case Diagram

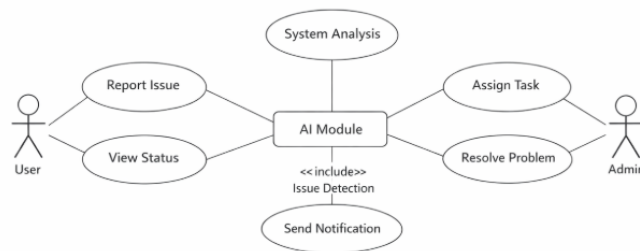


Figure 4.6: Use Case Diagram

This use case diagram shows how users and admins interact with the Smart Campus Problem Detection System. Users can report issues and view their status, while admins assign tasks and resolve problems. The AI module supports these actions by analyzing issues and sending notifications.

4.3.5 Sequence Diagram

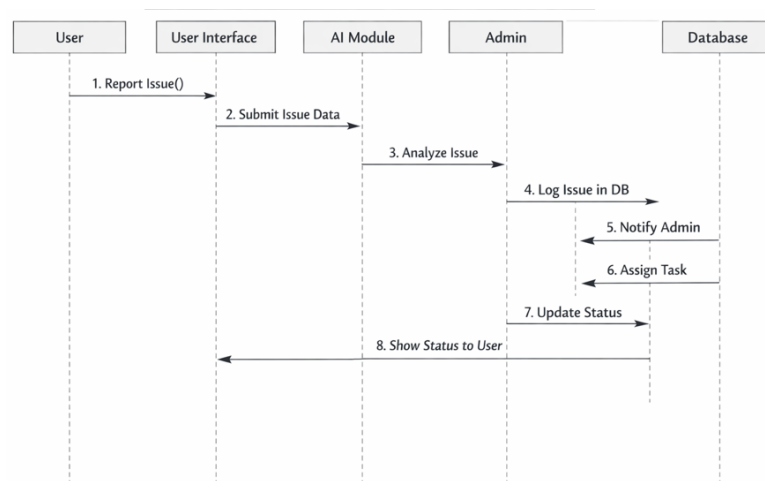


Figure 4.7: Sequence Diagram

This sequence diagram shows the step-by-step interaction between the user, system, AI module, admin, and database. It illustrates how an issue is reported, analyzed, stored, and assigned for resolution. Finally, the system updates and displays the issue status back to the user.

4.3.6 Activity Diagram

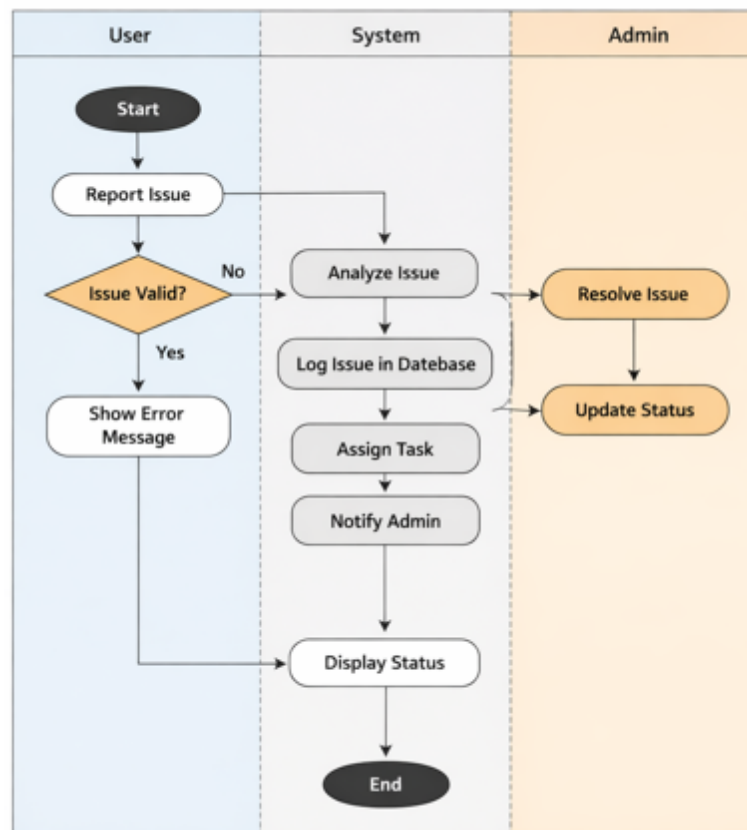


Figure 4.8: Activity Diagram

This activity diagram shows the step-by-step workflow of the Smart Campus system. It illustrates how an issue is reported, processed by the system, and resolved by the admin. It also highlights decision points and the overall flow from start to end.

4.3.7 Component Diagram

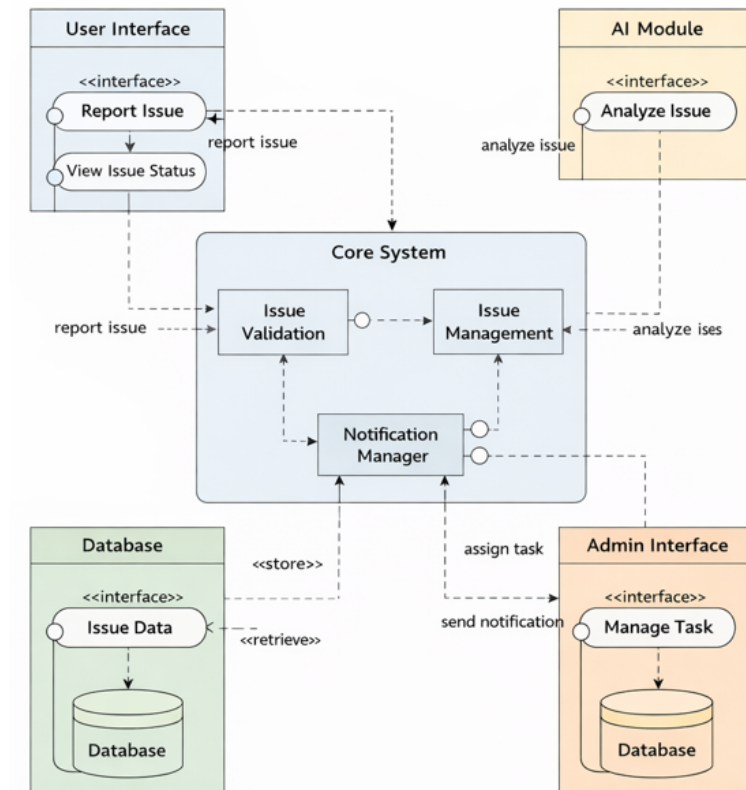


Figure 4.9: Component Diagram

This component diagram shows the overall architecture of the Smart Campus system by dividing it into major components like User Interface, AI Module, Database, and Admin Interface. It illustrates how these components interact with the core system to process and manage issues. The diagram helps in understanding the system's structure and how different modules work together.

4.3.8 Deployment Diagram

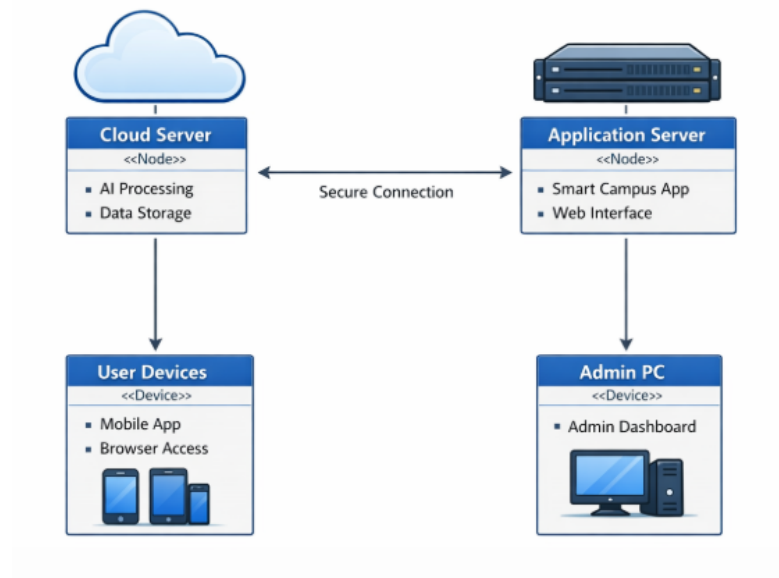


Figure 4.10: Deployment Diagram

This deployment diagram represents the physical setup of the Smart Campus Problem Detection System. It shows how components like the cloud server, application server, and database are deployed and connected through secure communication. Users access the system through mobile or web devices, while admins use a dedicated dashboard. The diagram helps in understanding how the system runs and interacts in a real-world environment.

CHAPTER 5

System Implementation

5.1 Modules Description

The Smart Campus Problem Detection and Resolution System is divided into several modules to ensure efficient functioning. Each module performs a specific task in the system.

User Module: This module allows users (students/faculty) to report campus issues through the system. Users can submit problems using text or images and also view the status of their complaints. It provides an easy interface for interaction.

AI-Based Problem Detection Module: This module analyzes the reported issues using AI techniques such as image processing and natural language processing. It identifies the type of problem and categorizes it automatically for faster resolution.

Database Module: This module stores all the information including user details, reported issues, and their status. It ensures proper data management and retrieval for future reference.

Admin Module: The admin monitors all reported issues and assigns tasks to the concerned department. The admin can update the status of issues and ensure that problems are resolved efficiently.

Notification Module: This module sends updates to users regarding the status of their reported issues. Notifications are sent whenever there is a change in the issue status or resolution.

5.2 Technology Used

The Smart Campus Problem Detection and Resolution System utilizes advanced technologies including Artificial Intelligence, Retrieval-Augmented Generation (RAG), and web development frameworks to ensure efficient problem detection and resolution.

1. **Programming Language:** Python is used as the primary programming language for developing backend logic, AI modules, and data processing tasks.
2. **Retrieval-Augmented Generation (RAG):** A custom Retrieval-Augmented Generation (RAG) pipeline is implemented to enhance system intelligence. It retrieves relevant information from internally stored Standard Operating Procedures (SOPs) within the codebase and generates appropriate responses for problem resolution. No external Large Language Model (LLM) or vector database is used; retrieval is based on structured logic and predefined data.
3. **Backend Framework:** The backend is built using Python standard libraries instead of external frameworks. `http.server.BaseHTTPRequestHandler` is used to handle HTTP requests and responses, enabling a lightweight server implementation. Other libraries include `json` for data handling, `os` for file operations, `re` for pattern matching, and `urllib` for URL handling.
4. **Custom Modules:** The system includes several custom modules such as `ai_engine` for core AI logic, `gemini_service` for AI and data processing services, `rag_pipeline` for retrieval and response generation, and `chatbot_engine` for managing chatbot interactions.
5. **Frontend Technologies:** HTML, CSS, and JavaScript are used to design a responsive and user-friendly interface.
6. **Database:** Currently, Standard Operating Procedures (SOPs) and related data are stored directly within the codebase instead of using a dedicated

database system.

7. **Cloud and Storage:** Google Colab is used for model training and experimentation, while Google Drive is used for storing datasets and trained models. Firebase is used for cloud storage and real-time data handling.
8. **Version Control:** Git and GitHub are used for code management, version tracking, and collaboration.

5.3 Algorithms Used

1. **Rule-Based NLP Classification:** A keyword-matching algorithm is used to classify user complaints into predefined categories such as Infrastructure, Electrical, and Plumbing. The system analyzes input text, identifies relevant keywords using pattern matching techniques, and assigns the complaint to the appropriate category based on predefined rules.
2. **Priority Prediction:** A keyword-based urgency detection algorithm is implemented to assign priority levels (High, Medium, Low) to user complaints. The system evaluates the presence of urgency-related keywords and determines the priority level to ensure timely resolution of critical issues.

5.4 Coding

The Smart Campus Problem Detection and Resolution System is implemented using multiple modules, each handling specific functionalities such as user interaction, AI-based analysis, database management, and notifications. Among these, the core component is the backend AI engine, which is responsible for analyzing user-reported issues.

The main backend code implements a Natural Language Processing (NLP) based AI engine that classifies complaints into different categories such as Infrastructure, Electrical, Plumbing, and others. It also predicts the priority level

(High, Medium, Low) using keyword-based urgency detection and contextual analysis.

This backend module plays a crucial role in automating problem detection, reducing manual effort, and improving response time. It ensures that issues are categorized accurately and prioritized efficiently, enabling faster resolution by the administration.

The following code snippet represents the core backend implementation of the AI engine:

```
class SmartCampusAI:

    CATEGORY_RULES = {
        'Infrastructure': ['building', 'wall', 'roof', 'damage'],
        'Electrical': ['light', 'power', 'fan', 'wire'],
        'Plumbing': ['water', 'pipe', 'tap', 'leak'],
        'IT & Network': ['wifi', 'internet', 'server'],
        'Cleanliness': ['garbage', 'waste', 'dirty'],
        'Safety & Security': ['fire', 'hazard', 'security'],
        'Furniture': ['chair', 'desk', 'table'],
        'Academic': ['class', 'exam', 'faculty'],
        'Hostel': ['room', 'mess', 'warden'],
        'Transport': ['bus', 'vehicle', 'parking']
    }

    HIGH_KEYWORDS = ['urgent', 'emergency', 'fire', 'danger']
    MEDIUM_KEYWORDS = ['broken', 'not working', 'leaking']

    def classify(self, text):
        lower = text.lower()

        best_category = 'Infrastructure'
        best_score = 0

        for category, keywords in self.CATEGORY_RULES.items():
```

```

        score = sum(1 for keyword in keywords if keyword in lower)
        if score > best_score:
            best_score = score
            best_category = category

    return best_category

def predict_priority(self, text):
    lower = text.lower()

    if any(word in lower for word in self.HIGH_KEYWORDS):
        return 'High'
    elif any(word in lower for word in self.MEDIUM_KEYWORDS):
        return 'Medium'
    else:
        return 'Low'

```

The complete implementation includes advanced features such as confidence scoring, multi-word phrase detection, and summary generation for intelligent decision-making. :contentReference[oaicite:0]index=0

CHAPTER 6

System Testing

6.1 Testing Methodology

The system testing is carried out to ensure that the Smart Campus Problem Detection and Resolution System functions correctly, efficiently, and reliably under different conditions. Both manual testing and functional testing techniques are applied to validate the performance of individual modules and the complete system.

The testing process focuses on verifying various components such as user input handling, AI-based problem detection using NLP and image processing, database storage, and notification mechanisms. Special attention is given to checking the accuracy of issue classification, priority prediction, and response generation using the RAG-based model.

Additionally, multiple test scenarios are executed to evaluate system performance, robustness, and error handling capabilities. The system is tested with different types of inputs, including valid, invalid, and edge cases, to ensure stability and reliability in real-world conditions.

6.2 Testing under Various Stages

The system is tested at multiple stages to ensure proper functionality and integration of all components:

Unit Testing: Each module such as User Interface, AI Module, Database, and Notification System is tested individually to verify its correctness and functionality.

Integration Testing: The interaction between modules is tested to ensure smooth data flow between components such as UI $\rightarrow$ AI Module $\rightarrow$ Database $\rightarrow$ Admin $\rightarrow$ Notification System.

System Testing: The complete system is tested as a whole to vali-

date overall performance, functionality, and system behavior under different scenarios.

User Acceptance Testing (UAT): The system is tested from the user's perspective to ensure that it meets user requirements and provides a satisfactory experience.

Performance Testing: The system is evaluated for response time, scalability, and efficiency, especially during multiple user requests and large data inputs.

6.3 Test Cases

S.No	Test Case	Expected Output	Result
1	User submits issue	Issue recorded successfully in system	Pass
2	AI analyzes issue	Correct category and prediction generated	Pass
3	Priority prediction	Correct priority (High/Medium/Low) assigned	Pass
4	Admin assigns task	Task assigned to appropriate department	Pass
5	Admin resolves issue	Issue status updated in database	Pass
6	Status displayed in UI	User receives status update	Pass
7	Invalid input handling	System displays error message	Pass
8	Multiple user requests	System handles requests without failure	Pass

Table 6.1: Test Cases for Smart Campus System

CHAPTER 7

Results and Discussion

7.1 Introduction

This chapter presents the results obtained from the implementation of the proposed smart campus problem detection and resolution system and discusses its overall performance. The main purpose of this chapter is to evaluate how effectively the system achieves its objectives, such as automatic problem classification, prioritization, and generation of accurate, context-aware responses using Generative AI and Retrieval-Augmented Generation (RAG).

The results highlight the system's ability to understand user complaints, classify them into relevant categories, and provide suitable action plans by retrieving information from institutional data sources. The performance of the system is analyzed based on factors such as accuracy, efficiency, and response time. This helps in understanding how well the system performs in real-world scenarios.

This chapter is structured to first present the system outputs, followed by a detailed discussion of the results. It also includes an evaluation of the system's effectiveness compared to traditional methods. Finally, key observations and possible improvements are discussed to provide insights for future enhancements.

7.2 Overview of Results

This section presents the key results obtained from the implementation of the smart campus problem detection and resolution system. The system was tested with different types of complaints such as electrical faults, water leakages, and damaged equipment. The results show that the system can effectively understand user inputs, classify problems into appropriate categories, and generate meaningful action plans.

The Generative AI component successfully analyzes textual inputs, and

when combined with the Retrieval-Augmented Generation (RAG) mechanism, it produces accurate and context-aware responses by retrieving relevant institutional data. The classification and prioritization modules help in identifying critical issues and ensure timely resolution. The system also enables efficient tracking of complaints through the admin dashboard.

Overall, the results indicate that the proposed system improves response time, reduces manual effort, and enhances decision-making in campus maintenance management, making it a practical and effective solution for real-world applications. The system also shows efficient performance when handling multiple complaints at the same time without delays. It provides useful insights through analytics, helping administrators understand common issues and trends. The user-friendly interface makes it easy for users to submit and track complaints. Overall, the system proves to be scalable, reliable, and suitable for real-world smart campus applications.

7.3 Result Screenshots

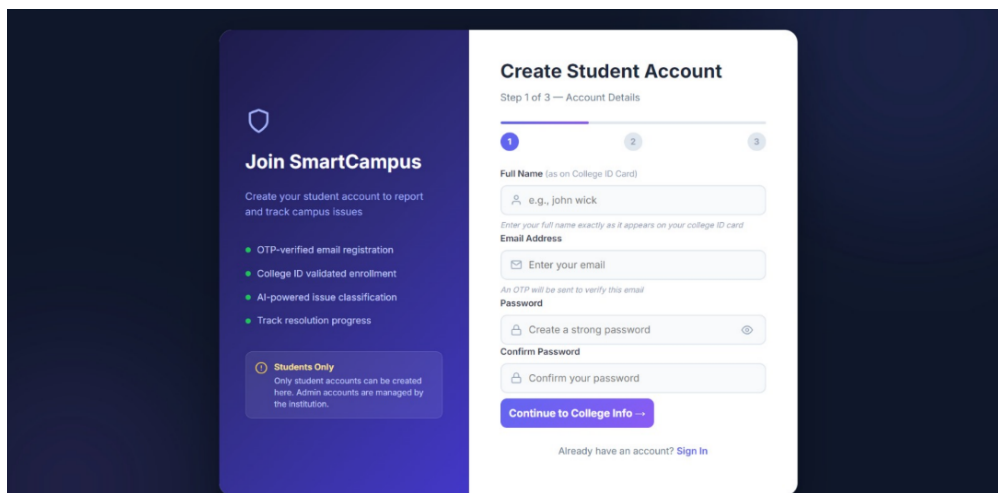
The image shows a user registration interface for 'SmartCampus'. On the left, a dark blue sidebar contains the 'Join SmartCampus' heading, a sub-heading 'Create your student account to report and track campus issues', a list of features (OTP-verified email registration, College ID validated enrollment, AI-powered issue classification, Track resolution progress), and a 'Students Only' note. The main area on the right is titled 'Create Student Account' and shows 'Step 1 of 3 — Account Details'. It includes a progress bar with three steps, input fields for 'Full Name', 'Email Address', 'Password', and 'Confirm Password', and a 'Continue to College Info' button. A 'Sign In' link is at the bottom.

Figure 7.1: User Registration Interface (Student Account Creation Page)

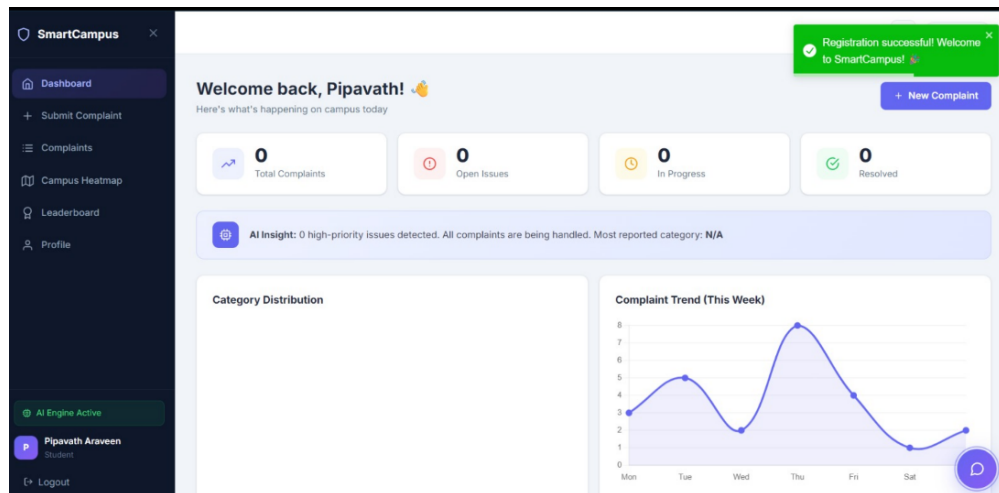


Figure 7.2: User Dashboard Showing Complaint Overview and AI Insights

SmartCampus

Submit New Complaint

Describe your issue and let AI classify and prioritize it automatically

Complaint Title

Water problem

13/100

Detailed Description

Water is not coming in bathroom

32/2000

Location

Block 1 - Ground Floor

Attach Image (optional)

AI Analysis

Enter your complaint description and click "Analyze with AI" to get automatic classification, priority prediction, and recommended action plan.

Student

AI Engine Active

Pipavath Araveen (Student)

Logout

Figure 7.3: Complaint Submission Interface with AI Analysis Panel

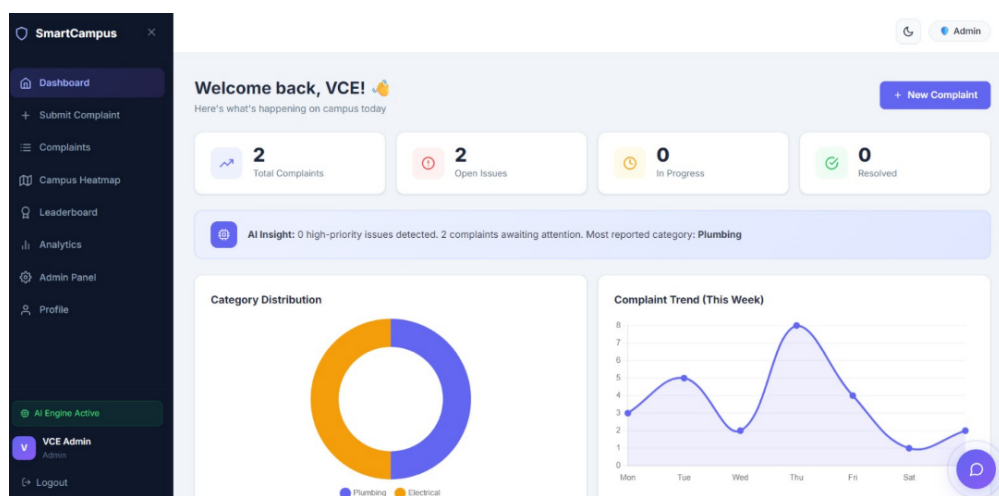


Figure 7.4: Admin Dashboard with Complaint Statistics and Category Distribution

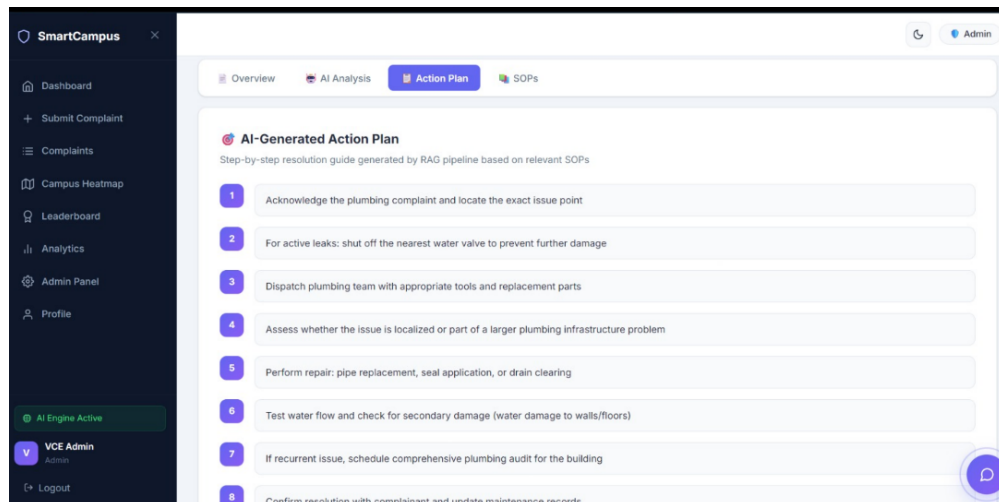


Figure 7.5: AI-Generated Action Plan for Issue Resolution

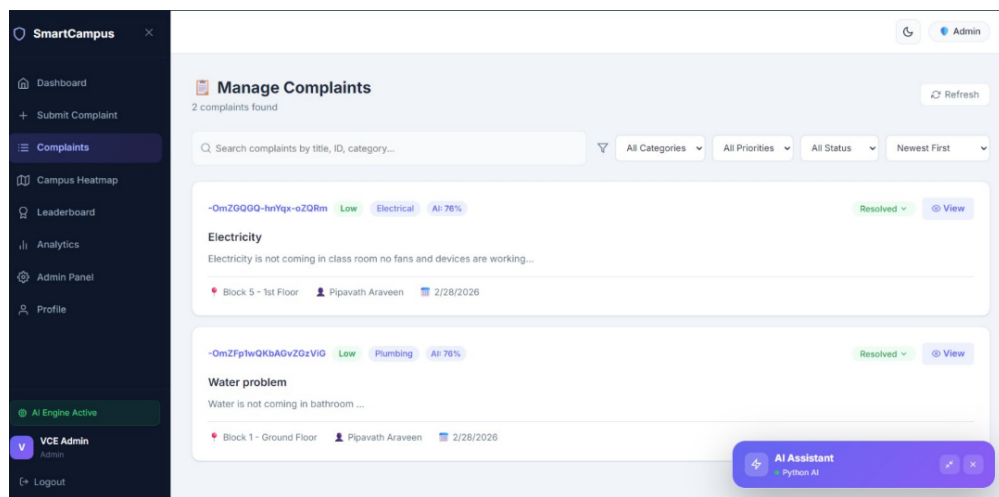


Figure 7.6: Complaint Management Interface for Admin



Figure 7.7: Analytics and Reports Dashboard Showing System Performance Metrics

7.4 Summary

This chapter presented the results and performance analysis of the proposed smart campus problem detection and resolution system. The findings show that the system effectively understands user complaints, classifies issues accurately, and generates relevant action plans using Generative AI and Retrieval-Augmented Generation (RAG).

The results demonstrate improvements in response time, accuracy, and overall efficiency compared to traditional systems. The integration of AI and RAG ensures context-aware and reliable outputs, while the prioritization and tracking features enhance decision-making and transparency.

Overall, the system proves to be a practical and scalable solution for managing campus infrastructure issues, contributing to a more efficient and intelligent smart campus environment.

CHAPTER 8

Conclusions and Future Scope

8.1 Conclusions

This project successfully developed an intelligent smart campus problem detection and resolution system using Generative Artificial Intelligence (AI) and Retrieval-Augmented Generation (RAG). The system addresses the limitations of traditional complaint management methods by providing an automated, efficient, and scalable solution for handling infrastructure-related issues in educational institutions.

The proposed system is capable of understanding user complaints, classifying them into appropriate categories, and assigning priority levels based on severity. By integrating RAG, the system generates accurate and context-aware responses using relevant institutional data such as maintenance records and policies. This ensures reliability and reduces the chances of incorrect outputs.

The project achieved its main objectives by improving response time, reducing manual effort, and enhancing transparency through complaint tracking and status updates. It also supports better decision-making for administrators by providing structured and organized information.

Overall, the system demonstrates the effective use of AI and RAG in solving real-world problems in campus management. It provides a strong foundation for building smarter, more efficient, and technology-driven campus environments.

8.2 Future Scope of Work

The proposed smart campus problem detection and resolution system demonstrates significant improvements over traditional methods; however, there are several opportunities for further enhancement and expansion.

One of the key areas for future work is the integration of real-time IoT sensors to automatically detect issues such as water leakage, temperature changes, or electrical faults without user intervention. This would make the system more proactive rather than reactive. Additionally, the system can be enhanced by incorporating image-based analysis using computer vision, allowing users to upload photos of issues for more accurate detection and classification.

Another important improvement is the use of advanced and fine-tuned language models specifically trained on institutional data. This would further improve the accuracy and relevance of responses generated by the system. The RAG mechanism can also be expanded by integrating larger and more dynamic data sources, such as live maintenance logs and real-time updates.

The system can be extended to include a mobile application for better accessibility and ease of use. Features such as push notifications, real-time tracking, and voice-based complaint submission can further improve user experience. Additionally, implementing multi-language support would make the system more inclusive and user-friendly.

From an administrative perspective, future work can include predictive analytics to identify recurring issues and prevent them in advance. This would help in better planning and resource allocation. Integration with other campus management systems can also create a more unified and efficient ecosystem.

Finally, the system can be scaled and adapted for use in other domains such as smart cities, hospitals, or corporate environments. By addressing these areas, the project can evolve into a more robust, intelligent, and widely applicable solution in the future.

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